

Peilin (Jack) Rao

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EDUCATION

Columbia University

Ph.D., Economics

New York, NY

Incoming, Sep 2026

University of California, Los Angeles (UCLA)

Bachelor of Science, Mathematics of Computation & Bachelor of Arts, Economics

Los Angeles, CA

Sep 2022 – Jun 2026

- **GPA:** 3.93/4.00 (Cum Laude)
- **Honors:**
 - ♦ **Undergraduate Research Fellows Program (URFP)** (Selected for \$3k grant, Winter 2026)
 - ♦ **Dean's Honors List (All eligible quarters)**
- **Selected Coursework:**
 - ♦ **Mathematics & Stats:** Real Analysis I & II, Linear Algebra, Probability Theory, Mathematical Statistics, Complex Analysis, Applied Numerical Methods, Optimization.
 - ♦ **Economics & Finance:** Micro/Macro Theory, Intro/Advanced Econometrics (+Lab), Data Science for Econ (+Lab), Economic Forecasting, Finance (+Lab), Financial Markets (+Lab), Investments (+Lab).

Working Papers

Rao, Peilin, and Randall R. Rojas. (2026). ["The Cost of Caution: Optimal Monetary Policy and the Value of Active Learning"](#)

Rao, Peilin, and Randall R. Rojas. (2025). ["Market Troughs as Bifurcation Events: A Liquidity Spiral Framework and Empirical Validation"](#)

RESEARCH EXPERIENCE

Undergraduate Research Fellow, URFP

UCLA

Advisor: Prof. Randall R. Rojas

Jan 2026 – Jun 2026

- Formulate a continuous-time dual control model of monetary policy under structural uncertainty to quantify the value of active learning versus the classic Attenuation Principle.
- Solve the non-linear Hamilton-Jacobi-Bellman (HJB) equation utilizing a high-fidelity Finite Difference Method (FDM) on a 125x125x125 tensor grid via JAX and GMRES iterative solvers.
- Identify a distinct "probing wedge" strategy that leverages induced economic volatility for accelerated model identification, achieving up to a 0.73 percentage point reduction in long-term inflation volatility.
- Participate in weekly 4-hour research workshops to refine research methodologies, present preliminary results, and conduct peer reviews of ongoing economic and social science research.

Teaching Experience

Reader (Teaching Assistant)

UCLA

Department of Economics

Sep 2025 – Jan 2026

- Course: Econ 144: Economic Forecasting (Instructor: Prof. Randall R. Rojas).

SKILLS

Dynamic Programming (HJB), Time Series Analysis (GARCH, State-Space Models), Panel Data Methods (DiD), Bayesian Econometrics, LaTeX.

Languages: Mandarin (Native), English (Fluent).